

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} G_x & G_y & G_z \end{bmatrix}$$

$$G_x = G_{px}^{(1)}$$

$$G_y = G_{py}^{(1)}$$

$$G_z = G_{pz}^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} G_{xxx} & G_{xxy} & G_{xxz} \\ G_{yyx} & G_{yyy} & G_{yyz} \\ G_{zzx} & G_{zzy} & G_{zzz} \\ G_{yzx} & G_{yyz} & G_{zzz} \\ G_{xxz} & G_{yzx} & G_{zzz} \\ G_{xxy} & G_{yyx} & G_{yzx} \end{bmatrix}$$

$$G_{xxx} = 2G_{fax}^{(1)} + 3G_{px}^{(1)}$$

$$G_{xxy} = -G_{fay}^{(1)} - G_{fby}^{(1)} + G_{py}^{(1)}$$

$$G_{xxz} = -G_{faz}^{(1)} + G_{fbz}^{(1)} + G_{pz}^{(1)}$$

$$G_{yyx} = -G_{fax}^{(1)} + G_{fbx}^{(1)} + G_{px}^{(1)}$$

$$G_{yyy} = 2G_{fay}^{(1)} + 3G_{py}^{(1)}$$

$$G_{yyz} = -G_{faz}^{(1)} - G_{fbz}^{(1)} + G_{pz}^{(1)}$$

$$G_{zzx} = -G_{fax}^{(1)} - G_{fbx}^{(1)} + G_{px}^{(1)}$$

$$G_{zzy} = -G_{fay}^{(1)} + G_{fby}^{(1)} + G_{py}^{(1)}$$

$$G_{zzz} = 2G_{faz}^{(1)} + 3G_{pz}^{(1)}$$

$$G_{yzx} = G_{f3}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxy} & G_{xxz} \\ G_{yyx} & 0 & G_{yyz} \\ G_{zzx} & G_{zzy} & 0 \\ G_{yzx} & -\frac{G_{yyz}}{2} & -\frac{G_{zzz}}{2} \\ -\frac{G_{xxx}}{2} & G_{zxy} & -\frac{G_{zzx}}{2} \\ -\frac{G_{xxy}}{2} & -\frac{G_{yyx}}{2} & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{xxy} = 2G_{py}^{(2)} - 2Q_{dxz}^{(1)}$$

$$G_{xxz} = 2G_{pz}^{(2)} + 2Q_{dxy}^{(1)}$$

$$G_{yyx} = 2G_{px}^{(2)} + 2Q_{dyz}^{(1)}$$

$$G_{yyz} = 2G_{pz}^{(2)} - 2Q_{dxy}^{(1)}$$

$$G_{zzx} = 2G_{px}^{(2)} - 2Q_{dyz}^{(1)}$$

$$G_{zzy} = 2G_{py}^{(2)} + 2Q_{dxz}^{(1)}$$

$$G_{yzx} = 3Q_{du}^{(1)} + Q_{dv}^{(1)}$$

$$\frac{G_{yyz}}{2} = G_{pz}^{(2)} - Q_{dxy}^{(1)}$$

$$\frac{G_{zzy}}{2} = G_{py}^{(2)} + Q_{dxz}^{(1)}$$

$$\frac{G_{xxz}}{2} = G_{pz}^{(2)} + Q_{dxy}^{(1)}$$

$$G_{zxy} = -3Q_{du}^{(1)} + Q_{dv}^{(1)}$$

$$\frac{G_{zzx}}{2} = G_{px}^{(2)} - Q_{dyz}^{(1)}$$

$$\frac{G_{xxy}}{2} = G_{py}^{(2)} - Q_{dxz}^{(1)}$$

$$\frac{G_{yyx}}{2} = G_{px}^{(2)} + Q_{dyz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & G_{yzy} & G_{yzz} \\ G_{zxx} & G_{zxy} & G_{zxz} \\ G_{xyx} & G_{xyy} & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = -Q_{du}^{(2)} + Q_{dv}^{(2)} + Q_s^{(1)}$$

$$G_{yzy} = G_{pz}^{(3)} + Q_{dxy}^{(2)}$$

$$G_{yzz} = -G_{py}^{(3)} + Q_{dxz}^{(2)}$$

$$G_{zxx} = -G_{pz}^{(3)} + Q_{dxy}^{(2)}$$

$$G_{zxy} = -Q_{du}^{(2)} - Q_{dv}^{(2)} + Q_s^{(1)}$$

$$G_{zzx} = G_{px}^{(3)} + Q_{dyz}^{(2)}$$

$$G_{xyx} = G_{py}^{(3)} + Q_{dxz}^{(2)}$$

$$G_{xyy} = -G_{px}^{(3)} + Q_{dyz}^{(2)}$$

$$G_{xyz} = 2Q_{du}^{(2)} + Q_s^{(1)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).